

Curriculum Vitae

Name	Jihye Yun	
Country	Korea, Republic of	
Current Affiliation	University of Ulsan College of Medicine, Asan Medical Center	
Department	Radiology	

Education Background

Ph.D., School of Computer Science and Engineering, Seoul National University, Seoul, Korea (2017)
M.S., School of Computer Science and Engineering, Seoul National University, Seoul, Korea (2011)
B.S., Department of Media, Soongsil University, Seoul, Korea (2005)

Professional Experience

Special Scholar, Department of Radiology, Asan Medical Center, Seoul, Korea (2019-)
Postdoctoral Research Fellow, University of Ulsan College of Medicine, Asan Medical Center, Seoul, Korea (2017-2019)

Fellowships

- The Scientific Research Fund of the Korean Society of Magnetic Resonance in Medicine
March 2022 - December 2022
Project Title: Deep learning-based lymphatic vessel detection in MR lymphangiography
- Fostering Next-generation Researchers Program
June 2018 - December 2019
Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education
Project Title: Implementation and Clinical Validation of Medical Image Net Using Whole-Body CT of Emergency Patients

Main Scientific Publications

- Jooae Choe, Hye Jeon Hwang*, Joon Beom Seo, Sang Min Lee, Jihye Yun, Min-Ju Kim, Jewon Jeong, Youngsoo Lee, Kiok Jin, Rohee Park, Jihoon Kim, Howook Jeon, Namkug Kim, Jaeyoun Yi, Donghoon Yu, Byeongsoo Kim, "Content-based image retrieval by using deep learning for interstitial lung disease with chest CT," *Radiology*, Jan. 2022.
- Sungchul Kim+, Sungman Cho+, Kyungjin Cho, Jiyeon Seo, Yujin Nam, Jooyoung Park, Kyuri Kim, Daeun Kim, Jeongeun Hwang, Jihye Yun, Miso Jang, Hyunna Lee, Namkug Kim*, "An open medical platform to share source code and various pre-trained weights for models to use in deep learning research," *Korean Journal of Radiology*, Dec. 2021.
- Young Hoon Cho, Joon Beom Seo*, Sang Min Lee, Namkug Kim, Jihye Yun, Jeong Eun Hwang, Jae Seung Lee, Yeon-Mok Oh, Sang Do Lee, Li-Cher Loh, Choo-Khoom Ong, "Radiomics approach for survival prediction in chronic obstructive pulmonary disease," *European Radiology*, Oct. 2021.
- Jihye Yun, Young Hoon Cho, Sang Min Lee, Jeongeun Hwang, Jae Seung Lee, Yeon-Mok Oh, Sang-Do Lee, Li-Cher Loh, Choo-Khoom Ong, Joon Beom Seo*, Namkug Kim*, "Deep radiomics-based survival prediction in patients with chronic obstructive pulmonary disease," *Scientific Reports*, Jul. 2021.
- Ji Eun Park+, Sungwon Ham+, Ho Sung Kim*, Seo Young Park, Jihye Yun, Hyunna Lee, Seung Hong Choi, Namkug Kim, "Diffusion and perfusion MRI radiomics obtained from deep learning segmentation provides reproducible and comparable diagnostic model to human in post-treatment glioblastoma," *European Radiology*, May 2021.
- Ho Yong Park, Hyun-Jin Bae, Gil-Sun Hong*, MinJee Kim, Jihye Yun, Sungwon Park, Won Jung Chung, Namkug Kim, "Realistic high-resolution body computed tomography image synthesis by using progressive growing adversarial network: visual turing test," *JMIR Medical Informatics*, Mar. 2021.
- Jung Su Lee+, Jihye Yun+, Sungwon Ham, Hyunjung Park, Hyunsu Lee, Jeongseok Kim, Jeong-Sik Byeon, Hwoon-Yong Jung, Namkug Kim*, Do Hoon Kim*, "Machine learning approach for differentiating cytomegalovirus esophagitis from herpes simplex virus esophagitis," *Scientific Reports*, Feb. 2021.
- Gil-Sun Hong, Kyung-Hyun Do*, A-Yeon Son, Kyung-Wook Jo, Kwang Pyo Kim, Jihye Yun, Choong Wook Lee, "Value of bone suppression software in chest radiographs for improving image quality and reducing radiation dose," *European Radiology*, Jan. 2021.
- Chong Hyun Suh+, Kyung Hwa Lee+, Young Jun Choi*, Sae Rom Chung, Jung Hwan Baek, Jeong Hyun Lee, Jihye Yun, Sungwon Ham, Namkug Kim*, "Oropharyngeal squamous cell carcinoma: radiomics machine-learning classifier from multiparametric MR images for determination of HPV infection status," *Scientific Reports*, Oct. 2020.
- Gwang Hyeon Choi+, Jihye Yun+, Jonggi Choi, Danbi Lee, Ju Hyun Shim, Han Chu Lee, Young-Hwa Chung, Yung Sang Lee, Beomhee Park, Namkug Kim*, Kang Mo Kim*, "Development of machine learning-based clinical decision support system for hepatocellular carcinoma," *Scientific Reports*, Sep. 2020.
- Jongha Park+, Jihye Yun+, Namkug Kim*, Beomhee Park, Yongwon Cho, Hee Jun Park, Mijeong Song, Minho Lee, Joon Beom Seo, "Fully automated lung lobe segmentation in volumetric chest CT with 3D U-net: validation with intra- and extra-datasets," *Journal of Digital Imaging*, Jan. 2020.
- Mingyu Kim, Jihye Yun, Yongwon Cho, Keewon Shin, Ryoungwoo Jang, Hyun-jin Bae, Namkug Kim*, "Deep learning in Medical Imaging," *Neurospine*, Dec. 2019.
- Jihye Yun, Ji Eun Park*, Hyunna Lee, Sungwon Ham, Namkug Kim, Ho Sung Kim, "Radiomic features and multilayer perceptron network classifier: a robust MRI classification strategy for distinguishing glioblastoma from primary central nervous system lymphoma," *Scientific Reports*, Apr. 2019.
- Sang Min Lee, Joon Beom Seo*, Jihye Yun, Young-Hoon Cho, Jens Vogel-Claussen, Mark L Schiebler, Warren B Geftter, Edwin JR Van Beek, Jin Mo Goo, Kyung Soo Lee, Hiroto Hatabu, James Gee, Namkug Kim, "Deep learning applications in chest radiography and computed tomography," *Journal of Thoracic Imaging*, Mar. 2019.
- Jihye Yun+, Jinkon Park+, Donghoon Yu, Jaeyoun Yi, Minho Lee, Hee Jun Park, June-Goo Lee, Joon Beom Seo, Namkug Kim*, "Improvement of fully automated airway segmentation on volumetric computed tomographic images using a 2.5 dimensional convolutional neural net," *Medical Image Analysis*, Jan. 2019.
- Jihye Yun, Yeo Koon Kim, Eun Ju Chun, Yeong-Gil Shin, Jeongjin Lee*, Bohyoung Kim*, "Stenosis map for volume visualization of constricted tubular structures: application to coronary artery stenosis," *Computer Methods and Programs in Biomedicine*, Feb. 2016.

- Hyunjoon Song, Jihye Yun, Bohyoung Kim*, Jinwook Seo*, "GazeVis: interactive 3D gaze visualization for contiguous cross-sectional medical images," IEEE Transactions on Visualization and Computer Graphics, Dec. 2013.
- Byeonghun Lee, Jihye Yun, Jinwook Seo, Byonghyo Shim, Yeong-Gil Shin, Bohyoung Kim*, "Fast high-quality volume ray casting with virtual sampling," IEEE Transactions on Visualization and Computer Graphics, Nov. 2010.

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